1. Business Scenario

BeautyCo is a cosmetics brand that plans to launch a new **digital marketing campaign** through platforms such as **TikTok, Instagram, and Shopee Live**.

To improve **product bundling and personalized marketing**, the brand wants to analyze:

“Which products are most frequently purchased together?”

These insights will help the company:

- Plan social media content
- Recommend product bundles
- Design targeted promotions
- Improve cross-selling strategies

In the transaction dataset:

- Each **row represents one customer's purchase transaction**
- Each **column represents a product**
- Value **1 = product purchased**
- Value **0 = product not purchased**

Objectives

- **Predictive analytics:**
  What additional products are customers likely to buy?
- **Prescriptive analytics:**
  If the company wants to increase sales of a specific product, **which other products should be recommended together?**

2. Data-Preprocessing

Tasks:

- Import data from a **CSV or Excel file**
- Study what **each row represents in the dataset**
- Select the **product columns** of interest
- Convert the dataset into **binomial format (true/false)**

Recommended operators in RapidMiner:

- **Select Attributes**
- **Set Role**
- **Numerical to Binominal**

(Equivalent Python steps may include pandas preprocessing.)

3. EDA

Explain the dataset structure:

- What does **each row represent**?

Explore the dataset:

- Which **products are most frequently purchased together?**
- Are there any **unexpected product combinations**?

4. Association Rule Modeling

Build association rules using the following steps:

1. Apply **FP-Growth**
    - Minimum Support = **0.2**
2. Apply **Create Association Rules**
    - Minimum Confidence = **0.7**

Questions

- How many **frequent itemsets** were discovered?
- Identify the **maximal frequent itemsets**.

5. Deployment and Interpretation

Tasks:

- Identify the **rule with the highest Confidence**
- Interpret the meaning of that rule
- Calculate:
    - **Support**
    - **Confidence**
    - **Lift**
- Suggest how this rule can be used for **marketing strategy**, such as:
    - Product bundles
    - TikTok content ideas
    - Promotional campaigns

Submission Requirements

Students must submit:

1) Answers to questions 3-4-5 (Cosmetic_STD-ID.pdf) AND

2) Coding (Cosmetic_STD-ID.xml or Cosmetic_STD-ID.ipynb)